DSA0503 – Query Processing for Data Science

CO2 - AT3 – Debugging

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Question 1: SQLite Database Connection

Aim

To connect to an SQLite database and create a Student table using Python.

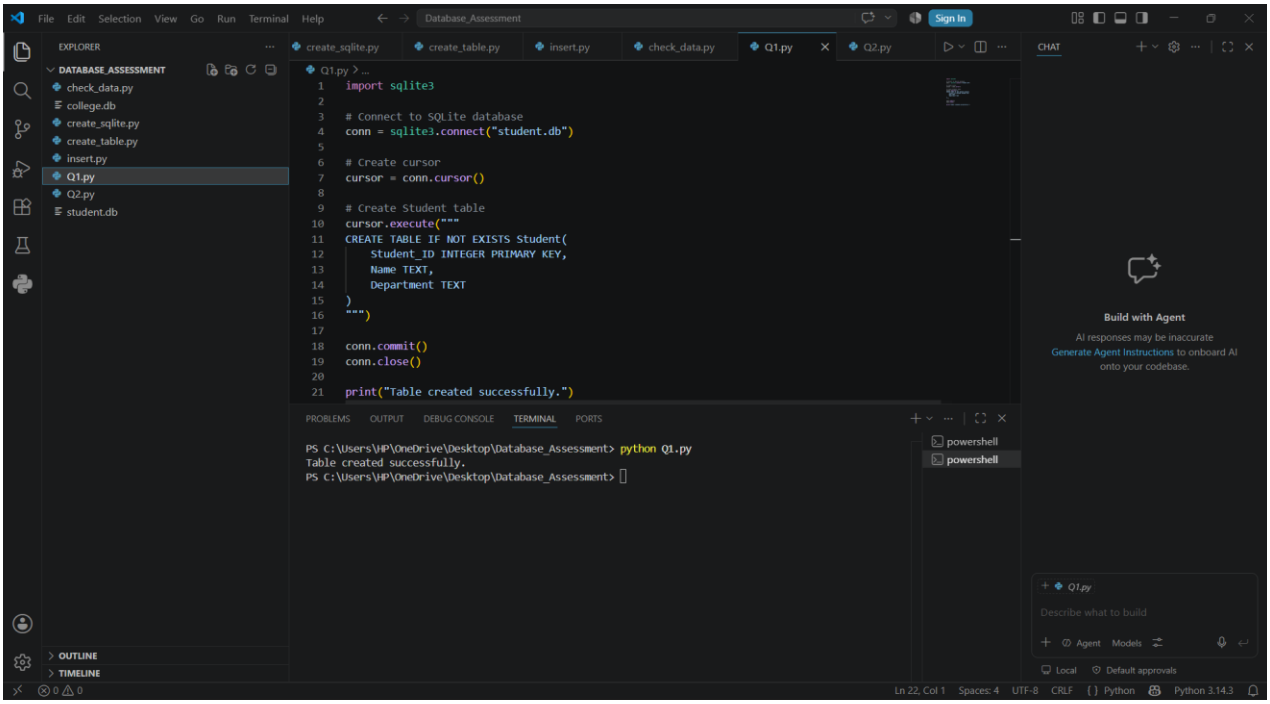
Software Required

- Python 3.x
- VS Code
- SQLite

Procedure

1. Open VS Code.
2. Connect to the SQLite database.
3. Create a cursor object.
4. Execute the SQL command to create the Student table.
5. Commit the changes.
6. Close the database connection.

Output



Result

The Student table was successfully created in the SQLite database.

Question 2: MySQL Record Insertion

Aim

To insert a new student record into a MySQL database using Python.

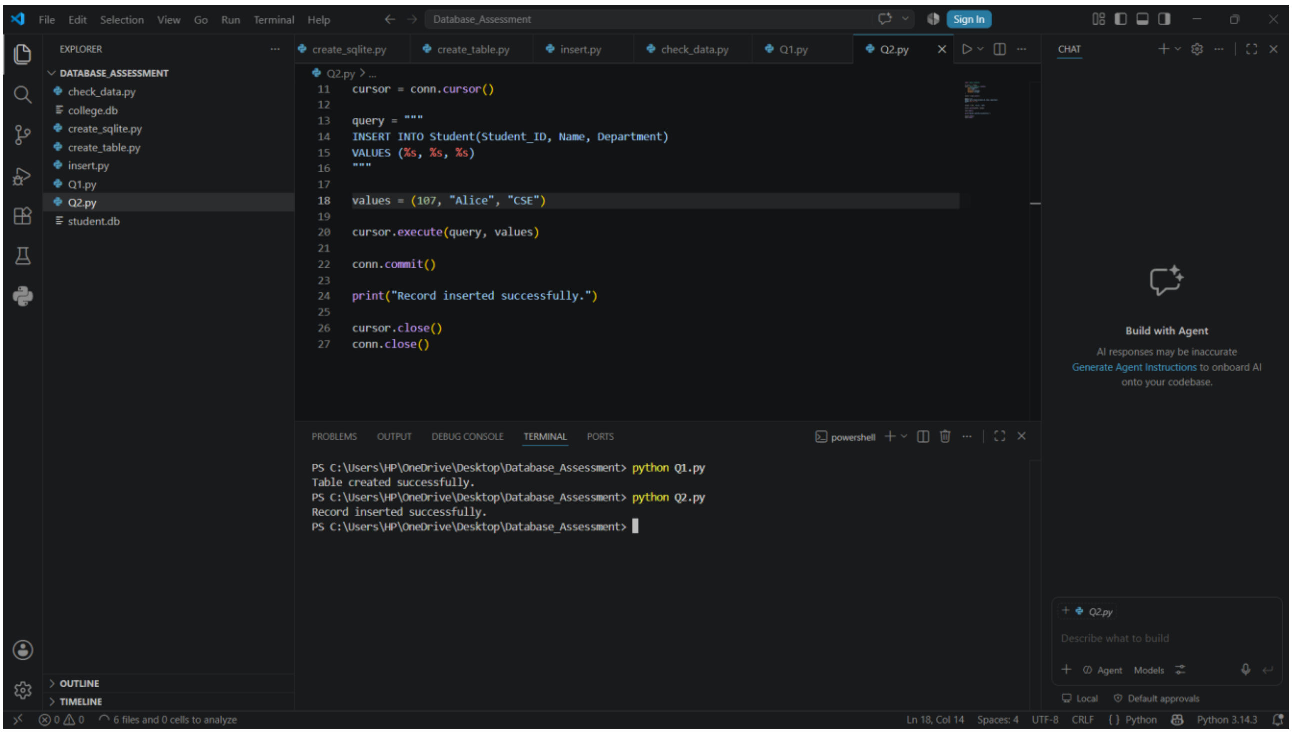
Software Required

- Python 3.x
- MySQL Server
- MySQL Workbench
- mysql-connector-python

Procedure

1. Connect to the MySQL database.
2. Create a cursor.
3. Execute the INSERT query.
4. Commit the transaction.
5. Close the connection.

Output



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Result

A new student record was successfully inserted into the Student table.

Question 3: PostgreSQL Data Retrieval

Aim

To retrieve all employee records from a PostgreSQL database.

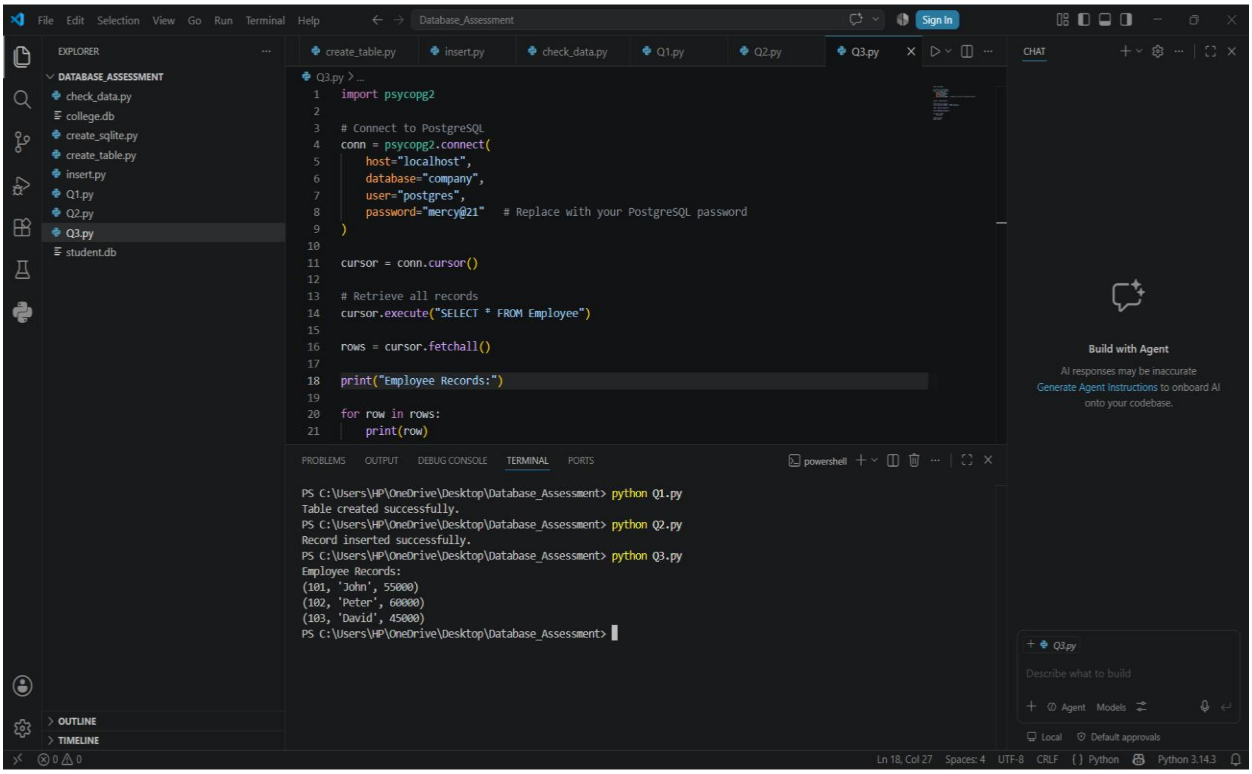
Software Required

- Python 3.x
- PostgreSQL
- pgAdmin
- psycopg2

Procedure

1. Connect to the PostgreSQL database.
2. Execute the SELECT query.
3. Fetch all records.
4. Display the records.
5. Close the connection.

Output



Result

All employee records were retrieved successfully.

Question 4: SQLite Record Update

Aim

To update the department of a student in the SQLite database.

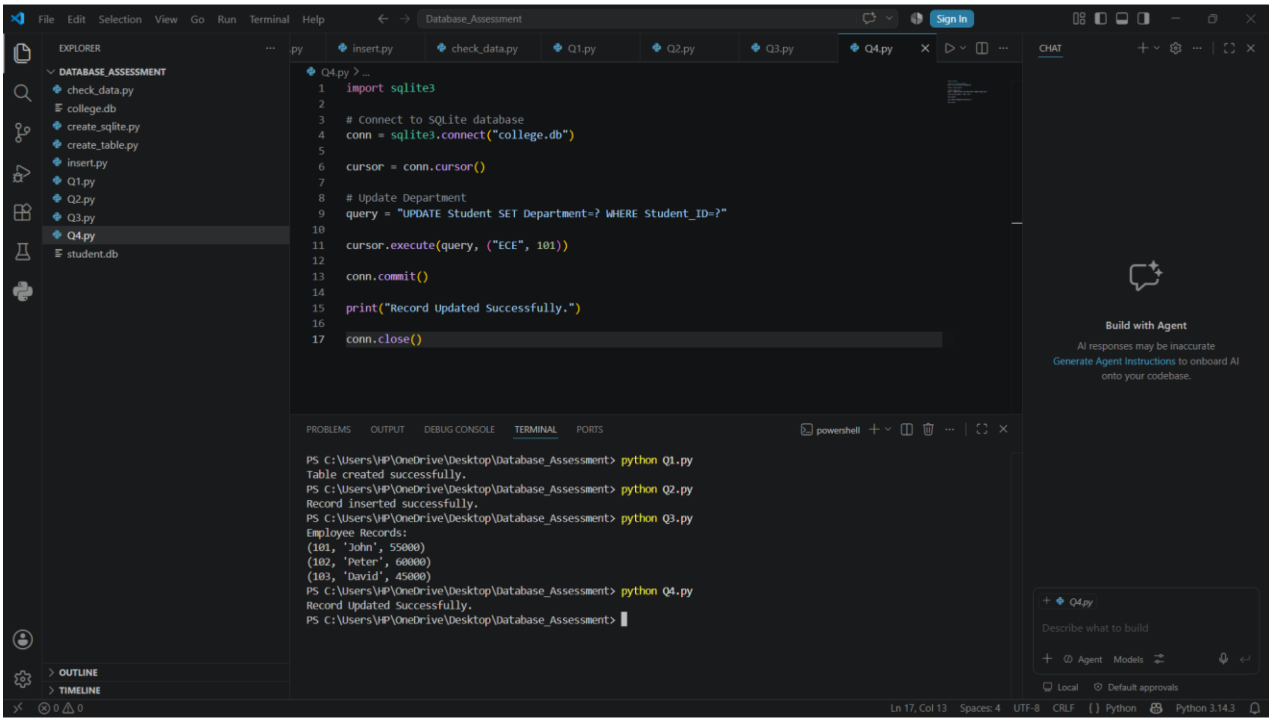
Software Required

- Python 3.x
- SQLite

Procedure

1. Connect to the SQLite database.
2. Execute the UPDATE query.
3. Commit the changes.
4. Close the connection.

Output



Result

The student's department was updated successfully.

Question 5: PostgreSQL Record Deletion

Aim

To delete a customer record from the PostgreSQL database.

Software Required

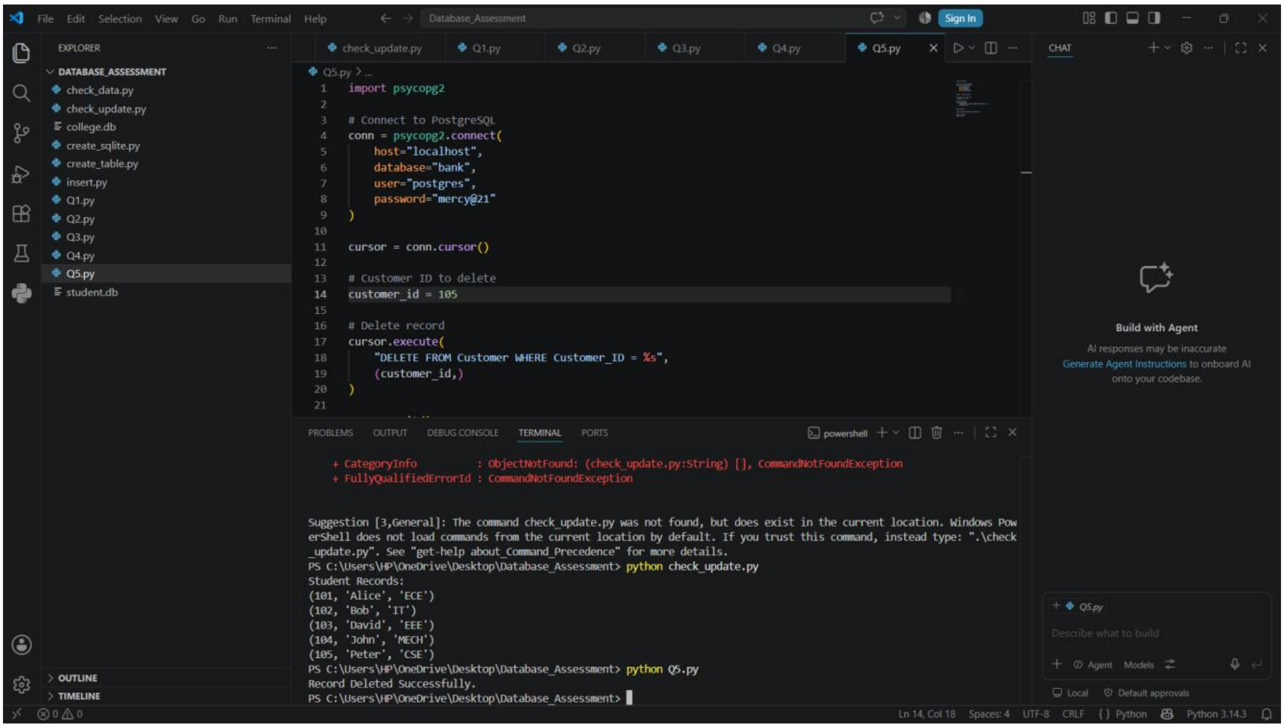
- Python 3.x
- PostgreSQL
- psycopg2

Procedure

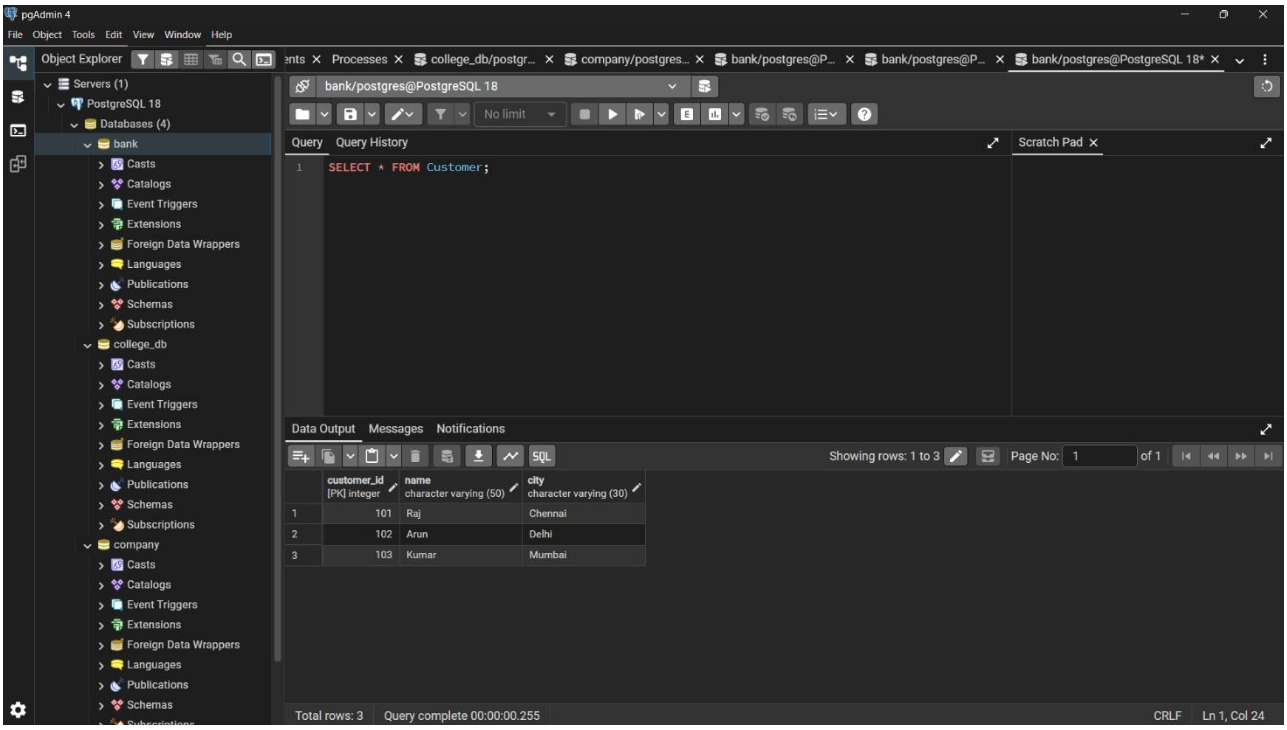
1. Connect to PostgreSQL.
2. Execute the DELETE query.
3. Commit the transaction.
4. Close the connection.

Output

Terminal:



PostgreSQL (Deleted):



Result

The selected customer record was deleted successfully.

Question 6: SQLite Record Retrieval

Aim

To retrieve a student record from the SQLite database using Student_ID.

Software Required

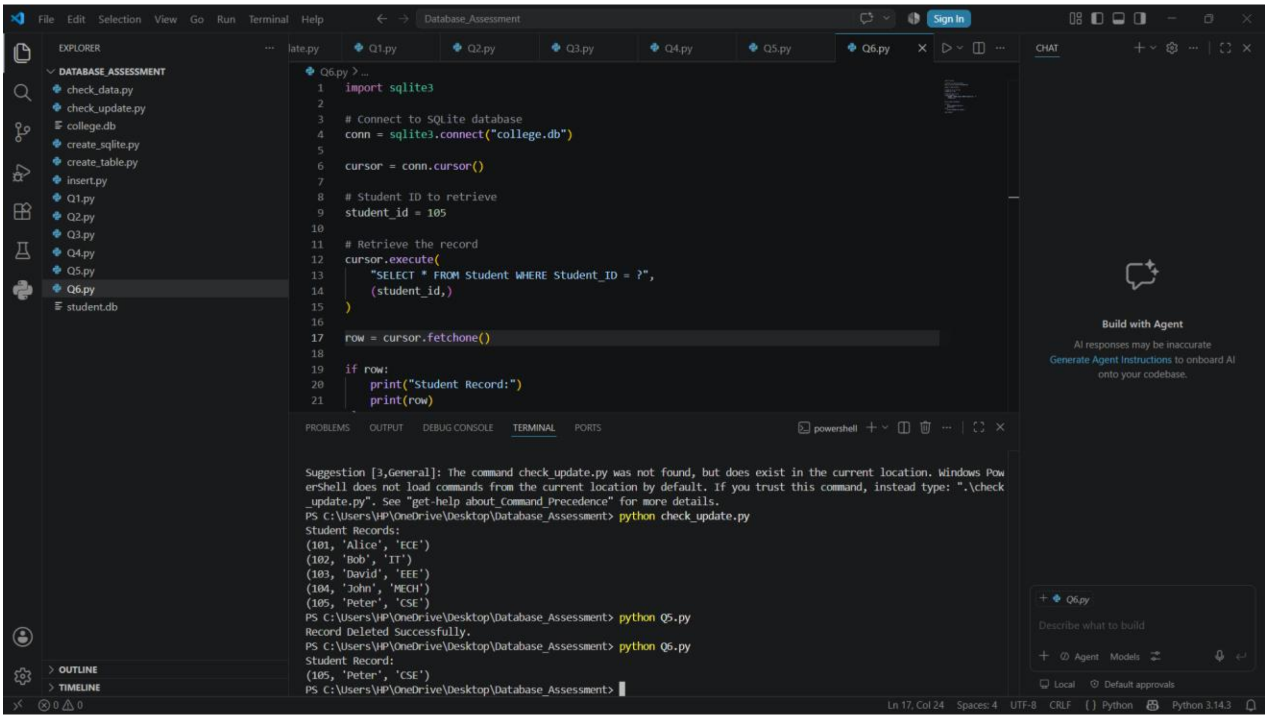
- Python 3.x
- SQLite

Procedure

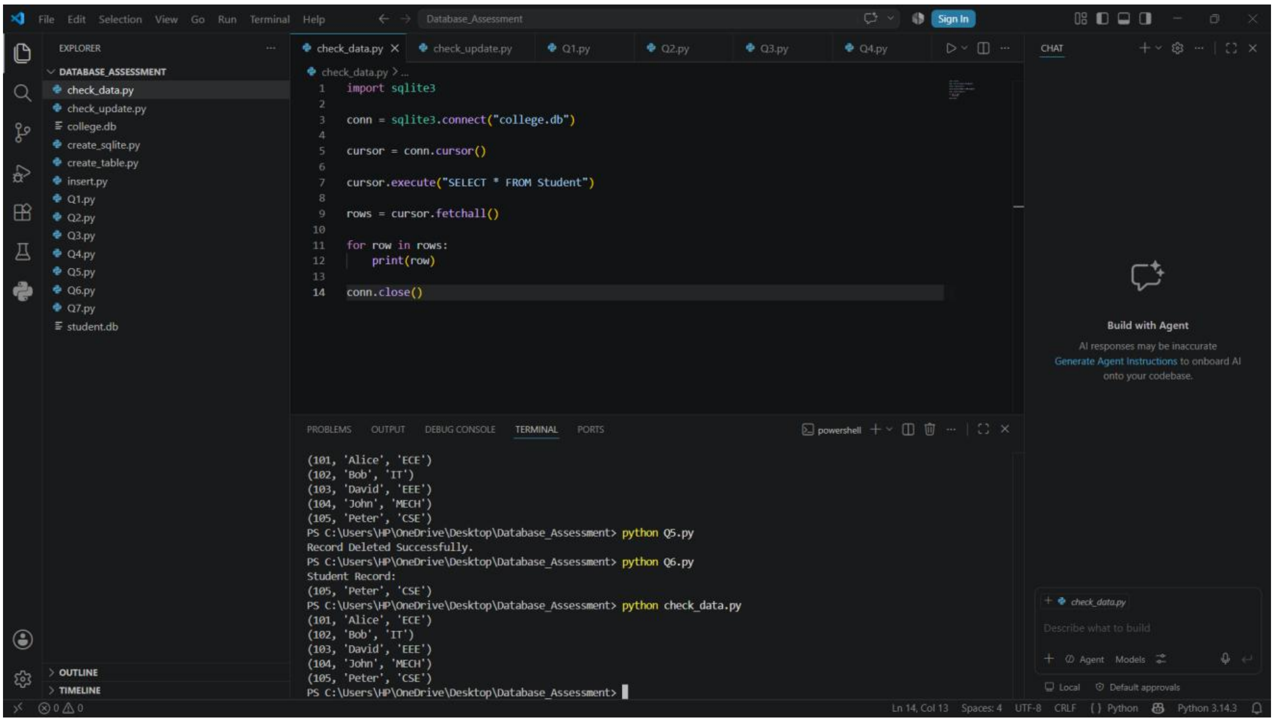
1. Connect to SQLite.
2. Execute the SELECT query.
3. Fetch the required record.
4. Display the record.
5. Close the connection.

Output

Terminal:



Retrieved:



Result

The required student record was retrieved successfully.

Question 7: MySQL Table Creation

Aim

To create a Student table in the MySQL database using Python.

Software Required

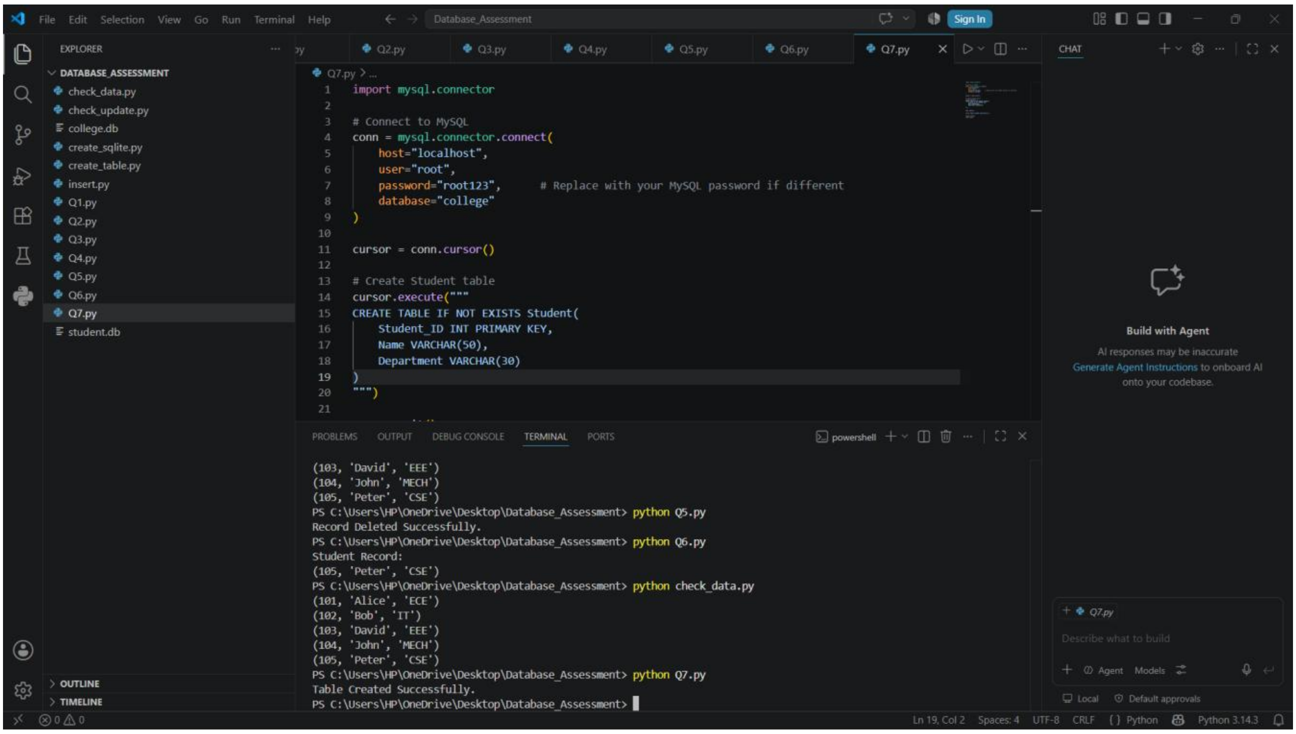
- Python 3.x
- MySQL Server
- mysql-connector-python

Procedure

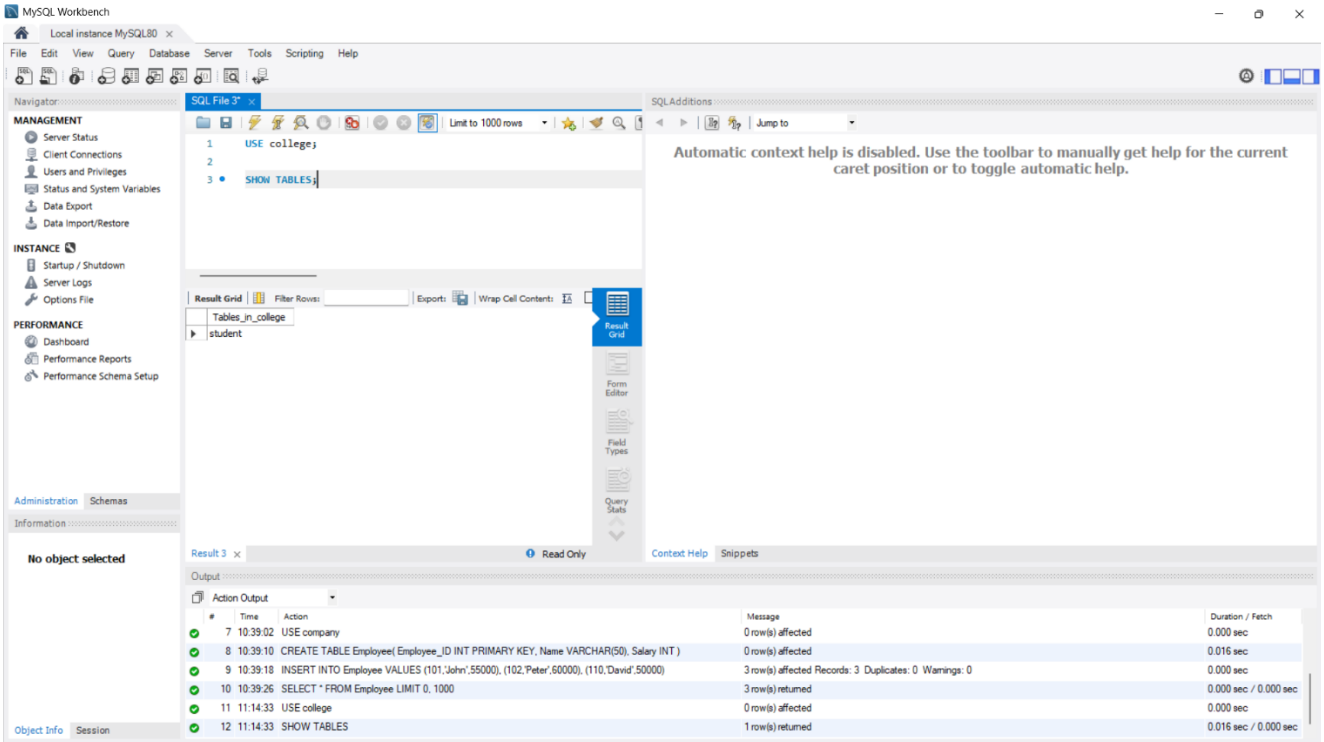
1. Connect to MySQL.
2. Execute the CREATE TABLE statement.
3. Commit the changes.
4. Close the connection.

Output

Terminal:



MySQL (Created Table):



Result

The Student table was created successfully in the MySQL database.

Question 8: PostgreSQL Record Insertion

Aim

To insert a new employee record into a PostgreSQL database.

Software Required

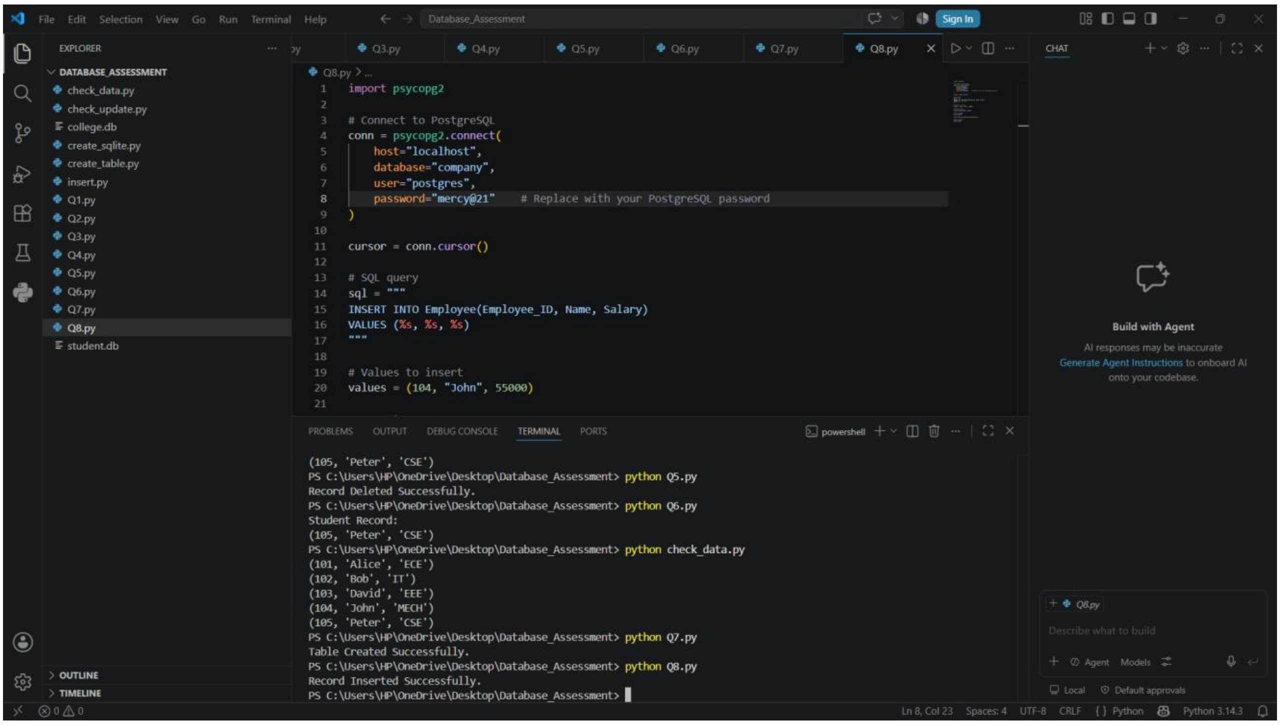
- Python 3.x
- PostgreSQL
- psycopg2

Procedure

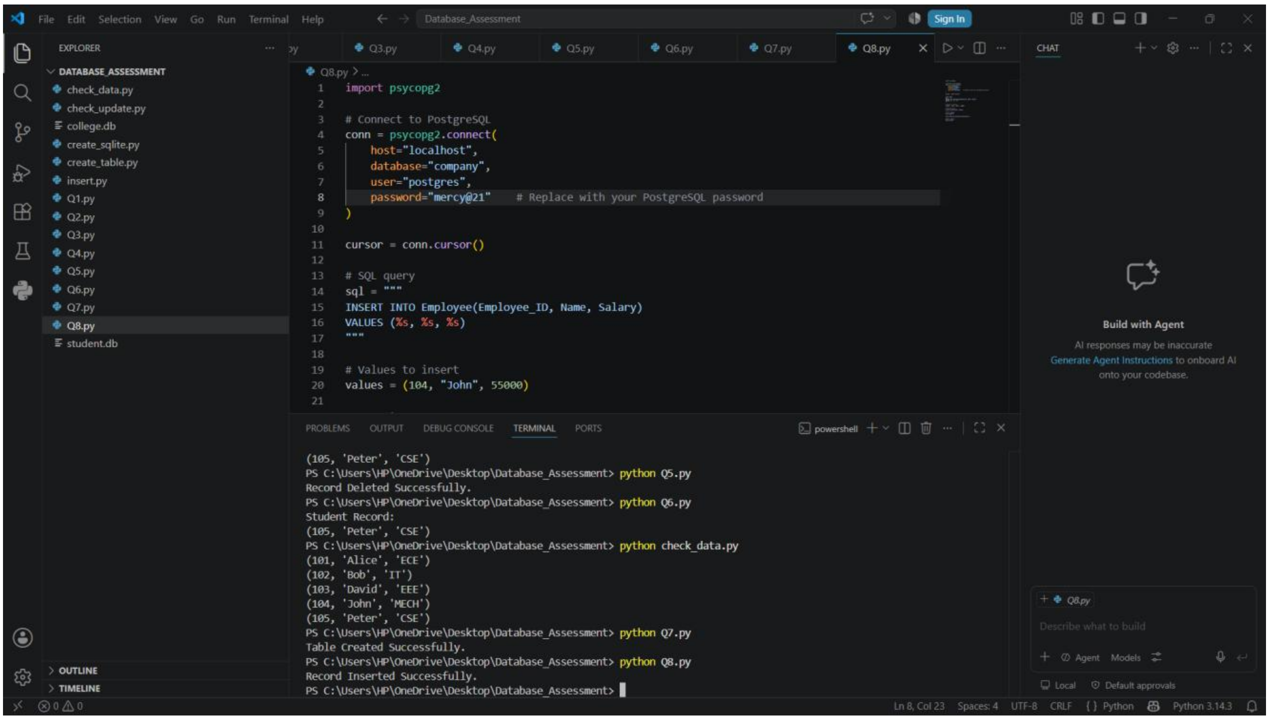
1. Connect to PostgreSQL.
2. Execute the INSERT query.
3. Commit the transaction.
4. Close the connection.

Output

Terminal:



PostgreSQL (Inserted Record):



Result

The employee record was inserted successfully into the Employee table.

Question 9: MySQL Record Update

Aim

To update the salary of an employee in the MySQL database.

Software Required

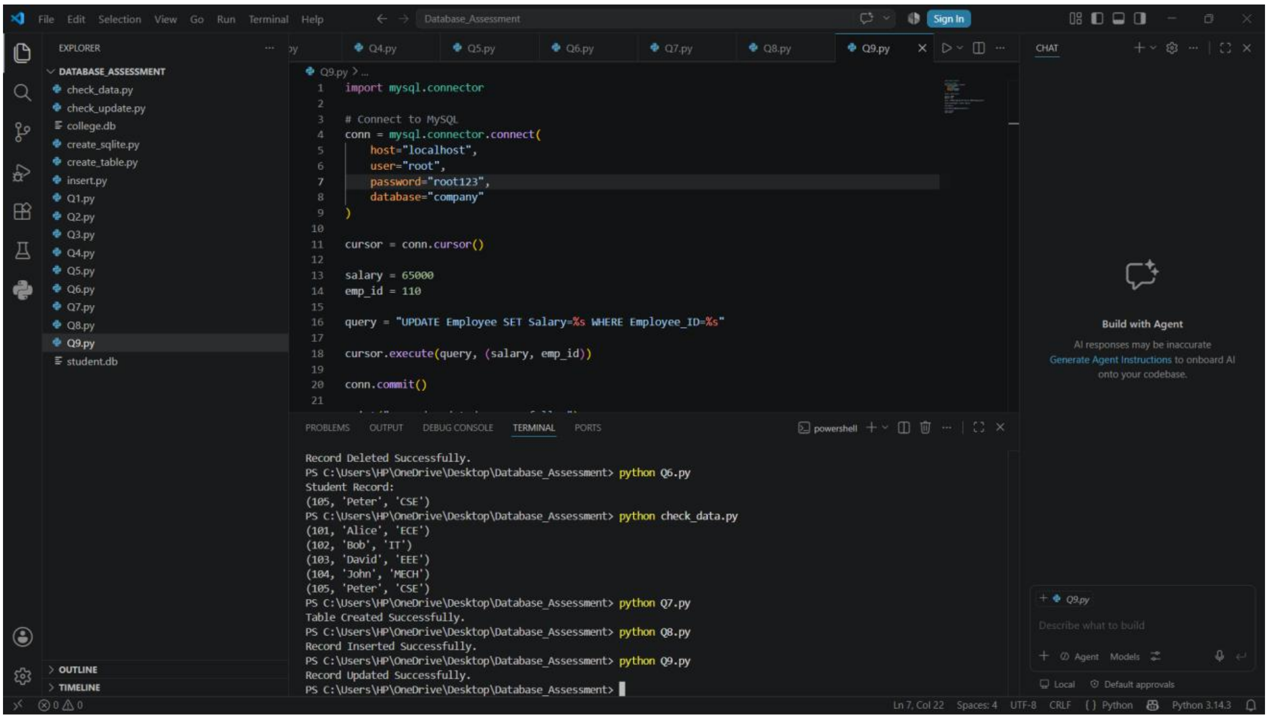
- Python 3.x
- MySQL Server
- mysql-connector-python

Procedure

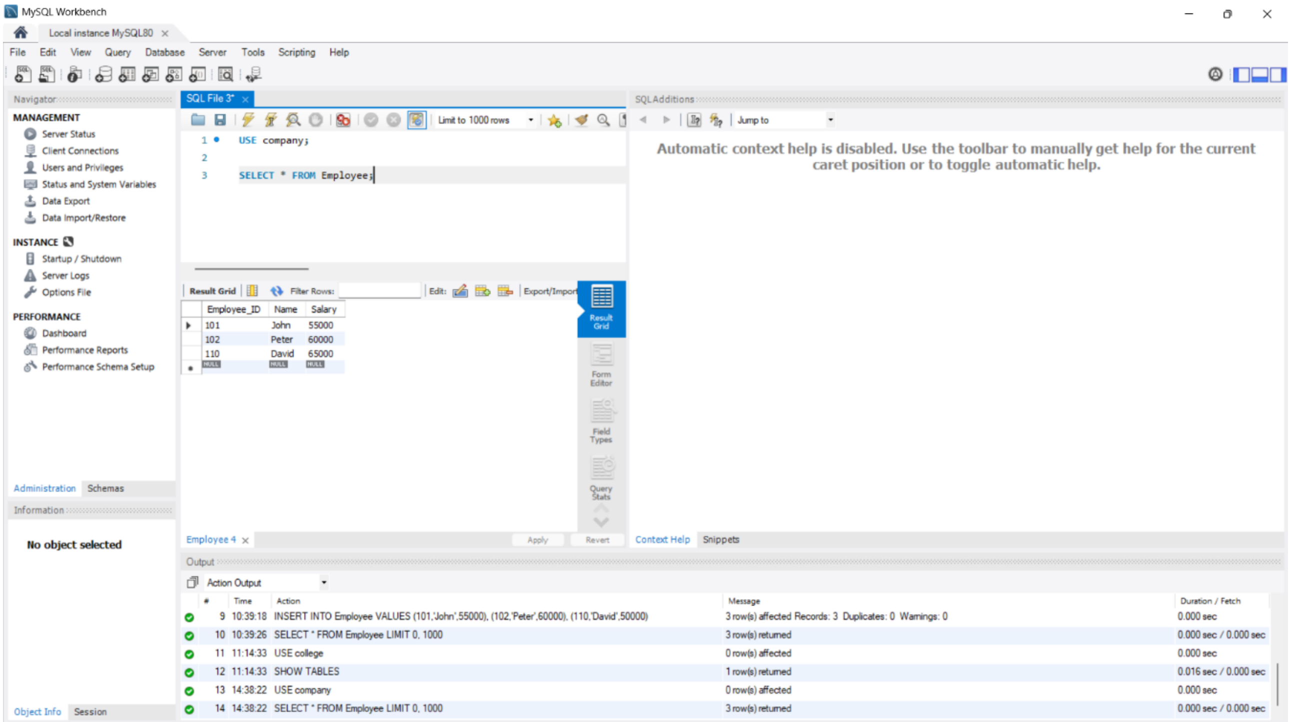
1. Connect to MySQL.
2. Execute the UPDATE query.
3. Commit the transaction.
4. Close the connection.

Output

Terminal:



MySQL (Updated Record):



Result

The employee salary was updated successfully.

Question 10: PostgreSQL Record Deletion

Aim

To delete all customer records belonging to Chennai from the PostgreSQL database.

Software Required

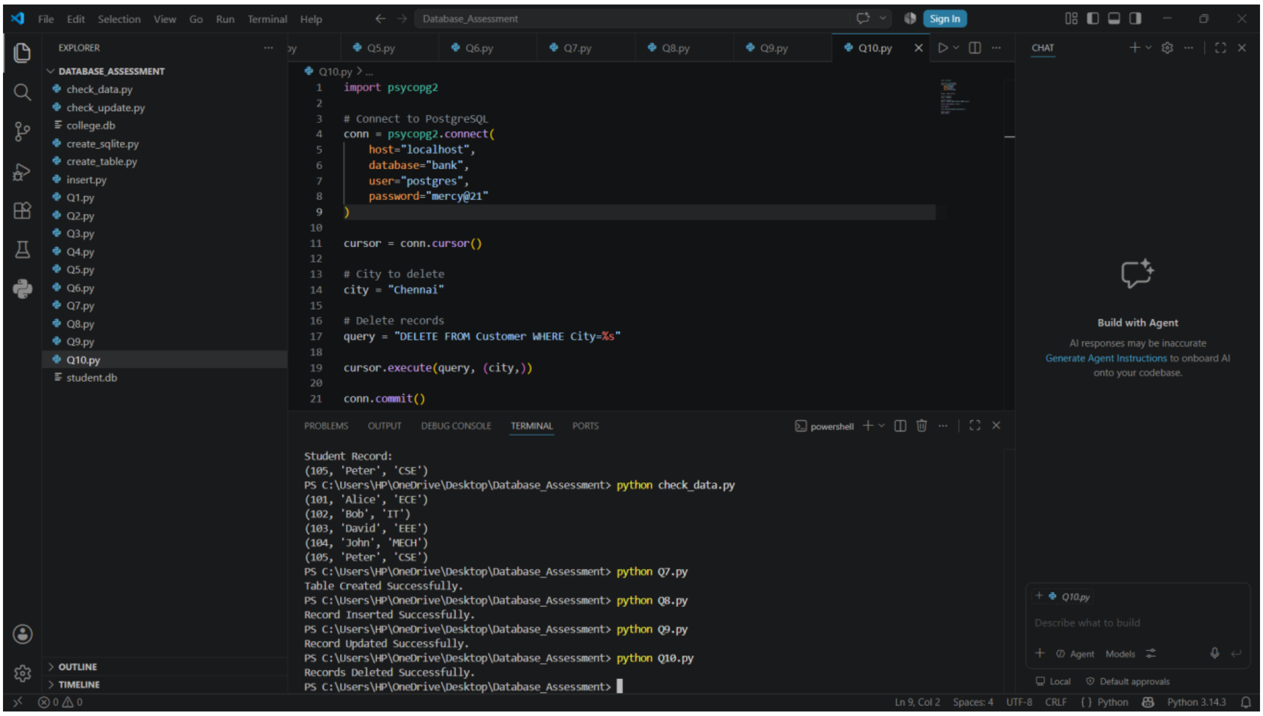
- Python 3.x
- PostgreSQL
- psycopg2

Procedure

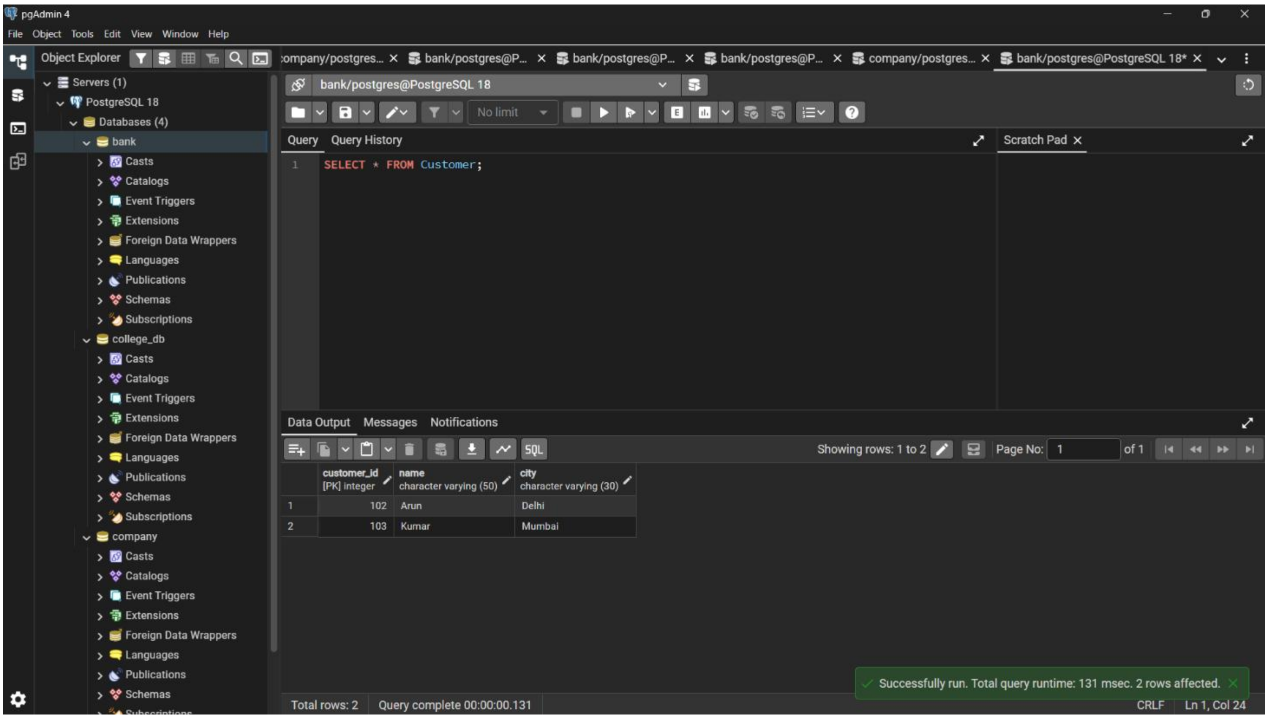
1. Connect to PostgreSQL.
2. Execute the DELETE query.
3. Commit the transaction.
4. Close the connection.

Output

Terminal:



PostgreSQL (Deleted Record):



Result

All customer records belonging to Chennai were deleted successfully.